

LENNART KANE

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Education

University of Waterloo

Bachelor of Applied Science Mechatronics Engineering

Sep 2025 – April 2030

Waterloo, Ontario

Relevant Coursework

- Data Structures
- Materials
- Computer-Aided Design
- Digital Computation
- Mechanics
- Chemistry
- Circuits
- Mathematics

Experience

Research and Development Intern

Jan 2026 – Apr 2026

KIRCHHOFF Automotive

Attendorn, Germany

- Designed and built a **weld current data acquisition** system using a **microcontroller**, **Hall effect sensor**, and a custom **binary serial protocol** to capture and log waveforms across welds.
- Implemented **weld failure detection** algorithms in C++, integrated with weld geometry models in **CATIA V5**, to **autonomously** assess weld **quality**.
- Programmed **FANUC** and **ABB** industrial robots using the **Teach Pendant** and **RAPID** language to **streamline** weld path operations and testing, cutting **labor costs** by **18%**.
- Developed a **camera trigger system** that processes weld current waveforms with **EMA filtering** to detect peaks and synchronize **high-speed video capture** of weld events.

Mechanical Lead

Sept 2022 – Aug 2025

Manning Robotics – FRC Team 4627

Calgary, AB

- Led **mechanical design** and **strategy** of competition robots over **3 seasons**, across **5 regional competitions**.
- Operated and trained team members on **machining**, **welding**, and **3D printing** equipment to fabricate custom robot components.
- Spearheaded **concept ideation** each season by leading brainstorming sessions and rapidly **prototyping** early designs to validate solutions before full build.
- Won the **Excellence in Engineering** award (2024) and **Autonomous** award (2023).

Sales Associate

Jan 2025 – Aug 2025

Sporting Life

Calgary, AB

- Consistently sold over **\$500** per shift across **ski**, **snowboard**, **cycling**, and **outdoor** departments through technical product expertise and personalized customer engagement.
- Received consistent positive customer feedback on **product knowledge**, **friendliness**, and **helpfulness**.

Projects

Automated Farm Irrigation System | ESP32, LoRa, Raspberry Pi

April 2026 – Present

- Designed a distributed sensor network using **ESP32** microcontrollers with **LoRa** radios across **multiple remote nodes**, powered by **solar panels** and batteries for off-grid operation.
- Built a centralized **Raspberry Pi** hub with **web dashboards** for remote monitoring and watering control.

Sputter – Autonomous Pancake-Making Robot | SolidWorks, C++, 3D Printing

November 2025

- Designed and built a fully **autonomous** pancake-making robot that dispenses, cooks, flips, and plates pancakes with **92%** consistency in a **4-person team**.
- Developed a **pneumatic pressure system** for viscous batter delivery, featuring a custom 2L bottle cap with **O-ring** sealing and a regulated **20psi** air supply.
- Modeled and **3D printed** a fully custom **PLA** frame and **3-axis Cartesian gantry** in **SolidWorks**, prioritizing rigidity, modularity, and design freedom.

Electric Go Kart | Welding, Shop Tools, Fusion360

2018

- Designed and built electric go kart using **shop tools**, **Flux-Cored Arc Welding**, and **Fusion360**.

Technical Skills

Engineering Software: SolidWorks CAD/FEA, Fusion360, CATIA V5, OnShape

Engineering Skills: GD&T, DFM, Product Testing, Soldering, MIG Welding, Machine Shop Qualified

Programming: C++, MS Excel, Basic Python